



LAB REPORT

Course Code ICE 3146 **Course Title :** Microprocessor and Interfacing Laboratory

Lab Task No. : 09

Lab Task Topic: Arduino UNO Interfacing with LEDs.

No.	Evaluation Criteria with Marks	Put Tick (✓) Mark					Marks Obtained	Remarks
		Excellent	Good	Fair	Poor	Fail		
REPORT WRITING								
1	Theory & Objectives (2.5)							
2	Materials & Methods (5)							
3	Data Analysis & Discussion/Diagram (10)							
4	Precautions and Conclusion (2.5)							
5	Time Management (5)							
	TOTAL							
LAB PERFORMANCE								
1	Preparation for Lab (5)							
2	Experiment and Engagement (15)							
3	Communication (5)							
	TOTAL							
15 th July, 2026 Date of Experiment		25 th July, 2026 Date of Submission					_____ Teacher’s Signature	

Semester : Summer **Year :** 2026 **Level-Term :** L2-T2 **Section :** A2

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Title: Arduino UNO Interfacing with LEDs.

Abstract:

This experiment was performed using the Arduino UNO board and Arduino IDE software to understand the basic concept of microcontroller programming and hardware interfacing with a buzzer. The main objective of this experiment was to become familiar with the Arduino UNO, learn how to write and upload a simple program through Arduino IDE, and control a buzzer through a digital output pin. In this experiment, the buzzer was connected to digital pin 8 of the Arduino UNO, and a simple program was uploaded so that the buzzer remained ON for 3 seconds and OFF for 2 seconds repeatedly. As a result, the buzzer produced a repeating alert sound similar to a ship siren pattern. This experiment helped to understand digital output control, timing using delay, program uploading, and simple interfacing between Arduino UNO and an external output device.

Introduction (Theory):

Arduino UNO is a widely used microcontroller development board based on the ATmega328P microcontroller. It is commonly used in embedded systems and electronics laboratories because it is easy to program and can interface with a large number of external devices. The Arduino UNO contains digital input/output pins, analog input pins, power pins, and a USB interface through which a program can be uploaded from a computer by using Arduino IDE software.

Interfacing means connecting an external hardware component with the microcontroller in such a way that the controller can control or communicate with that device. A buzzer is an electronic output device that converts electrical signals into sound. It is widely used in alarm systems, warning indicators, timers, and notification circuits. When the buzzer is connected to a digital output pin of the Arduino UNO, the microcontroller can control it by sending HIGH and LOW logic signals. A HIGH signal turns the buzzer ON, and a LOW signal turns it OFF.

In this experiment, a buzzer was connected to the Arduino UNO and controlled by a simple program written in Arduino IDE. The pinMode() instruction was used to configure the buzzer pin as an output pin, the digitalWrite() instruction was used to turn the buzzer ON and OFF, and the delay() function was used to control the time duration of each state. Because the buzzer was kept ON for 3000 milliseconds and OFF for 2000 milliseconds, the output produced a repeating sound pattern similar to a ship siren. This experiment demonstrated the basic working principle of Arduino programming, digital output control, timing control, and simple hardware interfacing.

Material:

Hardware	Software
1. Arduino UNO board & USB cable 2. Buzzer 3. Breadboard	1. Arduino IDE

- | | |
|------------------------------------|--|
| 4. Jumper wires | |
| 5. Personal Computer (PC) / Laptop | |

Method:

Circuit Connection:

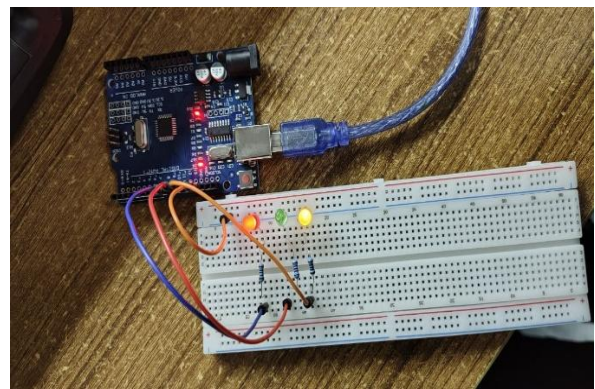
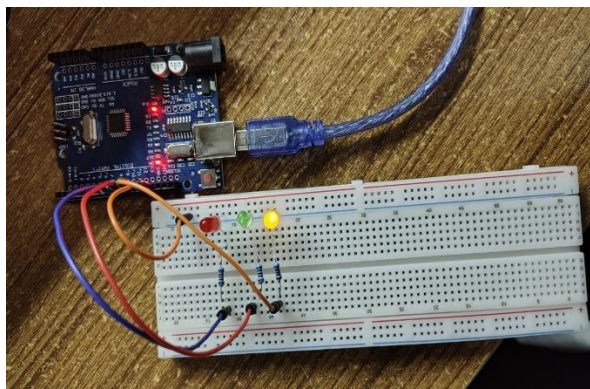
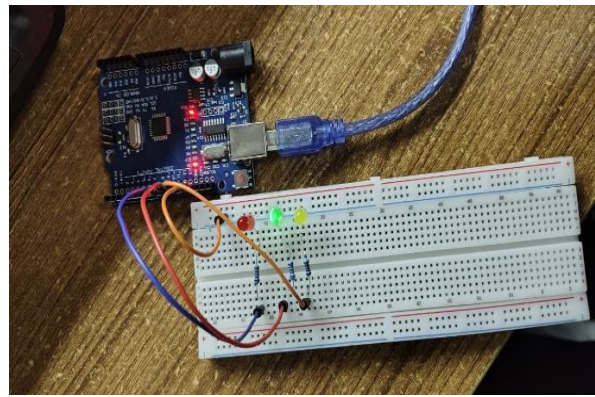
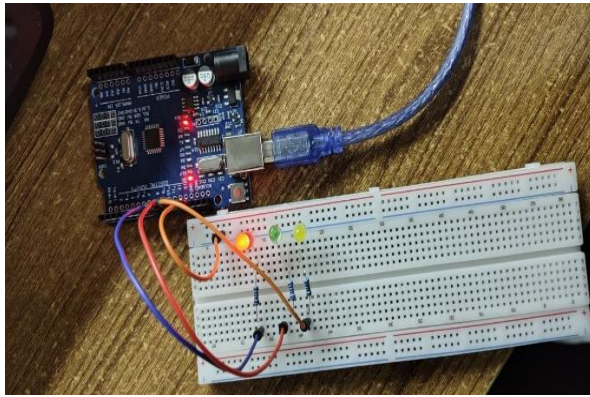


Fig 1: Circuit connection of Arduino UNO and LEDS

Code:

```
int redPin = 9;
int greenPin = 10;
int bluePin = 11;

void setup()
{
  pinMode(redPin, OUTPUT);
  pinMode(greenPin, OUTPUT);
  pinMode(bluePin, OUTPUT);
}

void loop()
{
```

```

// Red LED
analogWrite(redPin, 255);
analogWrite(greenPin, 0);
analogWrite(bluePin, 0);
delay(3000);

// Green LED
analogWrite(redPin, 0);
analogWrite(greenPin, 255);
delay(3000);

// Blue LED
analogWrite(greenPin, 0);
analogWrite(bluePin, 255);
delay(3000);

// Gradually increase red LED brightness
for (int brightness = 0; brightness <= 255; brightness++)
{
    analogWrite(redPin, brightness);
    delay(10);
}

// Gradually decrease red LED brightness
for (int brightness = 255; brightness >= 0; brightness--)
{
    analogWrite(redPin, brightness);
    delay(10);
}
}

```

Execution:

1. The red, green, and blue LEDs were placed on the breadboard.
2. One current-limiting resistor was connected in series with the positive terminal of each LED.
3. The positive terminal of the red LED was connected through a resistor to PWM pin 9.
4. The positive terminal of the green LED was connected through a resistor to PWM pin 10.
5. The positive terminal of the blue LED was connected through a resistor to PWM pin 11.
6. The negative terminals of all three LEDs were connected to Arduino GND.
7. The Arduino UNO was connected to the computer using a USB cable.
8. The program was written in the Arduino IDE.
9. Arduino UNO and the correct COM port were selected.
10. The program was verified and uploaded.
11. The red, green, and blue LEDs were activated sequentially for three seconds each.
12. The brightness of the red LED was gradually increased and decreased using PWM.

13. The complete sequence was repeated continuously..

Result:

Output Sequence:

Step	Red LED	Green LED	Blue LED	Duration / Condition
1	ON	OFF	OFF	3 Seconds
2	OFF	ON	OFF	3 Seconds
3	OFF	OFF	ON	3 Seconds
4	Gradually Dimmer	OFF	OFF	PWM value 0–255
5	Gradually Brighter	OFF	OFF	PWM value 255–0

Discussion:

In this experiment, three individual LEDs were interfaced with the Arduino UNO through PWM pins 9, 10, and 11. The `analogWrite()` function was used to control the LEDs by assigning PWM values between 0 and 255. A value of 255 produced maximum brightness, while a value of 0 turned the LED off. The supplied program uses these PWM-capable pins for red, green, and blue light control.

The red, green, and blue LEDs were activated sequentially for three seconds. The yellow colour combination was excluded from the experiment. Two for loops were then used to gradually increase and decrease the brightness of the red LED.

The observed outputs confirmed that the circuit connections, PWM pins, and program logic worked correctly. This experiment provided practical knowledge of LED interfacing, PWM output, brightness control, delay functions, and repeated execution using the Arduino UNO.